Lecture-22
Flow between plates

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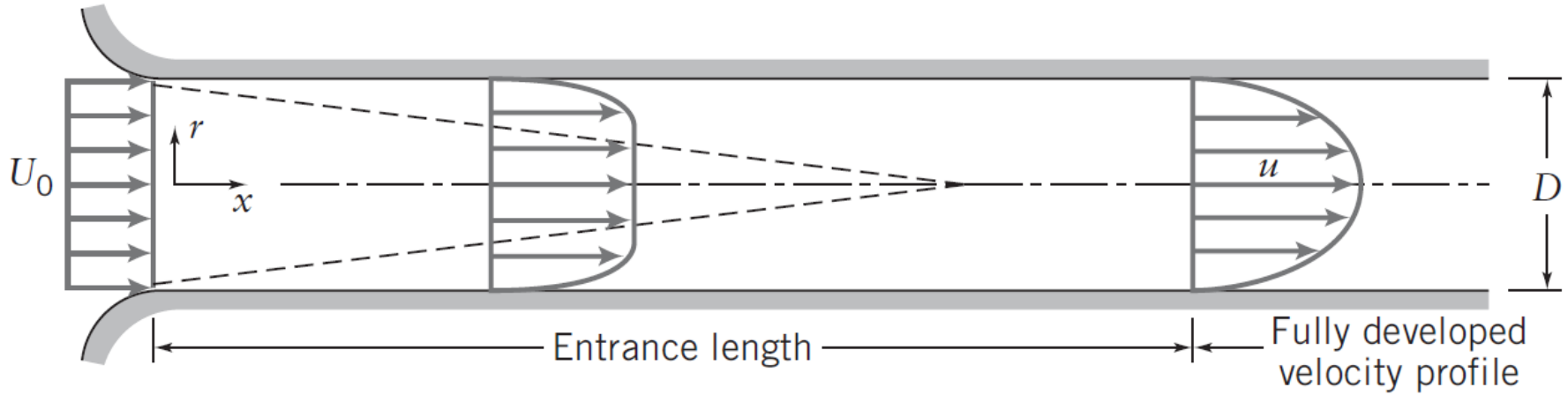
Learning Objectives

By the end of this lecture, you should be able to

1. Reduce the 3D NS equations by identifying key constraints like **steady-state**, **incompressible**, and **fully developed flow.**
2. Derive the velocity distribution for flow between plates, proving why **Poiseuille flow** yields a parabolic profile.
3. Calculate flow parameters such as **maximum velocity**, **volumetric flow rate**, and **wall shear stress** directly from the velocity profile.

Background information on internal flows

- Flows **completely bounded by solid surfaces** are called internal flows.
- For internal flows, the **classification into laminar and turbulent flows** is primarily a function of the **Reynolds number**.
- The distance downstream from the **entrance to the location at which fully developed flow begins** is called the **entrance length**.



Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Results for Poiseuille Flow

$$u = \frac{1}{2\mu}\left(\frac{dp}{dx}\right)y^2 - \frac{1}{2\mu}\left(\frac{dp}{dx}\right)ay = \frac{a^2}{2\mu}\left(\frac{dp}{dx}\right)\left[\left(\frac{y}{a}\right)^2 - \left(\frac{y}{a}\right)\right]$$

$$\tau_{yx} = \left(\frac{\partial p}{\partial x}\right)y + c_1 = \left(\frac{\partial p}{\partial x}\right)y - \frac{1}{2}\left(\frac{\partial p}{\partial x}\right)a = a\left(\frac{\partial p}{\partial x}\right)\left[\frac{y}{a} - \frac{1}{2}\right]$$

$$\frac{Q}{l} = -\frac{1}{12\mu}\left(\frac{\partial p}{\partial x}\right)a^3$$

$$\frac{Q}{l} = -\frac{1}{12\mu}\left[\frac{-\Delta p}{L}\right]a^3 = \frac{a^3\Delta p}{12\mu L}$$

$$\overline{V} = \frac{Q}{A} = -\frac{1}{12\mu}\left(\frac{\partial p}{\partial x}\right)\frac{a^3 l}{la} = -\frac{1}{12\mu}\left(\frac{\partial p}{\partial x}\right)a^2$$

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Revisiting the learning objectives

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